Tool-Augmented Question-Centric Retrieval Using QuIM-RAG for Reliable Knowledge-Grounded Question Answering

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***Abstract*—Retrieval-Augmented Generation (RAG) has emerged as an effective approach for improving the factual reliability of large language models (LLMs) by grounding generated responses in external knowledge sources. However, conventional RAG systems rely primarily on chunk-centric retrieval over text-only corpora, which often leads to semantic mismatch, information dilution, and hallucinations in real-world domain settings. To address these limitations, QuIM-RAG introduced a question-centric retrieval paradigm based on prototype-driven inverted indexing, improving retrieval precision and grounding compared to traditional RAG approaches. Nevertheless, QuIM-RAG assumes clean, pre-processed textual corpora and does not fully address challenges associated with heterogeneous web data and practical deployment environments.**

In this work, we extend the QuIM-RAG framework through data-centric and system-level enhancements designed for realistic web-based knowledge sources. Our approach incorporates tool-augmented corpus construction from heterogeneous data, including textual content, structured HTML tables, embedded images via optical character recognition (OCR), and linked PDF documents. The extracted information is normalized into a unified textual representation and integrated into a question-centric indexing pipeline using prototype-based inverted indexing. Additionally, we incorporate auxiliary components such as query rewriting and cross-encoder reranking to improve robustness under ambiguous queries, while clearly distinguishing these from core retrieval mechanisms.

We evaluate the proposed system on a medium-scale, domain-specific institutional corpus using BERTScore and RAG-oriented metrics, including faithfulness, answer relevance, context precision, and harmfulness, and compare it against lexical (BM25), dense, and hybrid retrieval baselines. Experimental results demonstrate that the proposed extensions achieve competitive semantic accuracy while improving grounding and reliability in domain-specific question answering. These findings highlight the importance of data-centric and system-level enhancements for enabling practical deployment of question-centric RAG frameworks in real-world environments.

***Index Terms*—Retrieval-Augmented Generation, Question-Centric Retrieval, QuIM-RAG, Web Knowledge Extraction, Domain-Specific Question Answering, Large Language Models**

I. Introduction

Large Language Models (LLMs) have demonstrated significant progress in natural language understanding and generation, enabling applications ranging from conversational assistants to domain-specific knowledge systems. Despite these advancements, LLMs remain fundamentally limited by their reliance on parametric knowledge acquired during pretraining, which may be outdated or incomplete. As a result, LLMs are prone to hallucinations, generating fluent but factually incorrect outputs, thereby limiting their reliability in real-world deployments [5], [6].

Retrieval-Augmented Generation (RAG) has been proposed as an effective strategy to mitigate hallucination by augmenting LLMs with external knowledge retrieval mechanisms. By integrating dense retrieval over document collections with sequence-to-sequence generation, RAG enables models to produce responses grounded in retrieved passages rather than relying solely on internal memory [1]. Since its introduction, numerous RAG-based systems have explored improved retrievers, hybrid lexical-semantic indexing, and advanced reranking methods to enhance retrieval quality and factual grounding.

Nevertheless, most existing RAG systems employ chunk-centric retrieval, where fixed-length document segments are directly matched against user queries in an embedding space. This approach introduces two key challenges. First, semantic mismatch between user queries and document chunks can result in irrelevant retrievals, particularly for underspecified or implicitly phrased queries. Second, as corpus size increases, chunk-centric retrieval becomes susceptible to information dilution, retrieving multiple loosely related passages that degrade downstream generation and increase hallucination risk [13].

To address these limitations, QuIM-RAG (Question-to-Question Inverted Matching) introduced a question-centric retrieval paradigm that shifts the retrieval unit from document chunks to automatically generated questions [2]. In this approach, representative questions are generated for each chunk, and user queries are matched against these questions in a shared embedding space. Furthermore, QuIM-RAG employs a prototype-based inverted index, clustering semantically similar questions into prototype vectors to enable efficient coarse-to-fine retrieval. This design improves retrieval precision and

reduces information dilution compared to conventional RAG systems.

While QuIM-RAG provides a strong conceptual foundation for improving retrieval reliability, it primarily assumes clean, pre-processed textual corpora and does not explicitly address challenges posed by real-world web data. In practical deployment scenarios, particularly in institutional and enterprise environments, relevant knowledge is often distributed across heterogeneous sources such as structured tables, embedded images, scanned documents, and linked PDF files. Conventional RAG and QuIM-RAG pipelines are not designed to directly ingest or effectively utilize such semi-structured and non-textual knowledge without substantial preprocessing.

In this work, we extend the QuIM-RAG framework toward real-world applicability through data-centric and system-level enhancements. We introduce a tool-augmented web corpus construction pipeline that integrates automated extraction of domain knowledge from diverse web artifacts, including textual content, structured tables, images via optical character recognition (OCR), and PDF documents. The extracted knowledge is normalized into a unified textual representation and incorporated into a question-centric indexing pipeline using prototype-based inverted indexing. Additionally, we incorporate auxiliary components such as query rewriting and cross-encoder reranking to improve retrieval robustness and precision under ambiguous queries.

We evaluate the proposed system on a medium-scale, domain-specific corpus using BERTScore and RAG-oriented metrics, including faithfulness, answer relevance, and context precision, and compare performance against lexical, dense, and hybrid retrieval baselines. Experimental results demonstrate that the proposed extensions achieve competitive semantic accuracy while improving grounding and reliability in domain-specific question answering. These findings highlight the importance of data-centric and system-level enhancements for enabling practical deployment of question-centric RAG frameworks in real-world environments.

II. Related Work

This section reviews prior research on retrieval-augmented generation, question-centric retrieval, web-based knowledge acquisition, and evaluation of hallucination in large language models. We position our work within this literature and clarify how it extends existing approaches.

A. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines parametric language modeling with non-parametric knowledge retrieval to improve factual grounding for knowledge-intensive tasks. The foundational RAG framework proposed by Lewis et al. integrates dense document retrieval with sequence-to-sequence generation, enabling large language models to produce responses conditioned on retrieved passages rather than relying solely on internal parameters [1]. This approach demonstrated strong performance on open-domain question answering and fact verification benchmarks, establishing RAG as a core paradigm for mitigating hallucinations in LLM outputs.

Subsequent research has explored improvements to the retrieval component through hybrid lexical-semantic retrieval, enhanced document encoders, and reranking mechanisms. Despite these advances, most RAG systems continue to treat document chunks as atomic retrieval units. This assumption becomes increasingly brittle in large or heterogeneous corpora, where chunk-level semantic alignment with user queries is often weak [13]. As a result, chunk-centric retrieval remains susceptible to information dilution, frequently retrieving loosely related contexts that degrade answer quality.

B. Question-Centric Retrieval and QuIM-RAG

To address the limitations of chunk-centric retrieval, recent work has explored question-centric retrieval paradigms, where retrieval units are derived from questions rather than raw document segments. QuIM-RAG introduced Question-to-Question Inverted Matching, generating representative questions for each document chunk and performing retrieval by matching user queries against these questions in a shared embedding space [2]. This approach improves semantic alignment between queries and indexed knowledge.

In addition, QuIM-RAG employs a prototype-based inverted index, clustering semantically similar questions into prototype vectors to enable efficient coarse-to-fine retrieval. This design reduces retrieval ambiguity and mitigates information dilution compared to conventional dense RAG pipelines. However, the original QuIM-RAG formulation primarily assumes clean, text-only corpora and does not address practical challenges associated with real-world web data, such as structured tables, visual content, and scanned documents.

C. Web-Based Knowledge Extraction and Domain Corpora

In real-world settings, domain-specific knowledge is often distributed across heterogeneous web artifacts, including structured HTML tables, embedded images, and linked documents such as PDFs. Traditional RAG pipelines largely ignore these sources or treat them as unstructured text, limiting knowledge coverage and retrieval effectiveness. Prior research on web-scale information extraction has explored techniques for table understanding, OCR-based text extraction, and document parsing; however, these methods are rarely integrated cohesively into RAG systems [11], [10].

Recent studies emphasize the importance of data-centric approaches for improving LLM reliability, particularly in enterprise and institutional environments where relevant knowledge is not confined to clean narrative text [21]. Nevertheless, existing RAG and question-centric retrieval frameworks typically rely on pre-curated datasets and do not provide end-to-end solutions for automated corpus construction from heterogeneous web sources. Our work addresses this gap by integrating tool-augmented web data extraction directly into a question-centric RAG pipeline.

D. Hallucination and Evaluation of RAG Systems

Hallucination remains a significant challenge in LLM-based systems, particularly when retrieval is insufficient or irrelevant. Prior studies have categorized hallucinations and proposed mitigation strategies, including retrieval augmentation, consistency verification, and faithfulness evaluation [5], [6], [14]. While retrieval-augmented models reduce hallucination, their effectiveness depends heavily on retrieval quality and contextual relevance.

Recent work has proposed evaluation metrics tailored to RAG systems. BERTScore evaluates semantic similarity between generated answers and reference texts [19]. Additionally, frameworks such as RAGAS introduce metrics including faithfulness, answer relevance, and context precision, providing a more comprehensive evaluation of retrieval-generation pipelines [4]. Our evaluation methodology builds upon these approaches by applying multi-metric evaluation across lexical, dense, hybrid, and question-centric retrieval baselines.

E. Positioning of Our Work

In contrast to prior work, our approach extends QuIM-RAG through data-centric and system-level enhancements aimed at practical deployment. Rather than proposing a new retrieval paradigm, we focus on enabling question-centric retrieval to operate effectively over heterogeneous web-based knowledge sources. By integrating tool-augmented corpus construction, prototype-based indexing within a unified pipeline, and comprehensive RAG-oriented evaluation, our work advances the applicability of question-centric RAG frameworks beyond controlled benchmark datasets and into realistic deployment environments.

III. System Overview

The proposed system extends the QuIM-RAG question-centric retrieval framework through data-centric and system-level enhancements aimed at real-world deployment. As illustrated in Fig. 1, the architecture is organized into three main stages: (i) tool-augmented web corpus construction, (ii) question-centric knowledge indexing, and (iii) query-time retrieval and answer generation. This modular design enables effective operation on heterogeneous real-world data while preserving the core principles of question-centric retrieval.

A. Tool-Augmented Web Corpus Construction

The first stage constructs a domain-specific knowledge base from real-world web sources. Unlike conventional RAG pipelines that assume pre-processed textual datasets, the proposed system directly ingests web content and applies multiple extraction techniques to capture heterogeneous information.

The pipeline performs domain-constrained web crawling, followed by extraction of visible textual content, structured HTML tables, embedded images, and linked PDF documents. Textual content is cleaned by removing boilerplate elements such as headers, navigation menus, and footers.

Structured tables are parsed and converted into textual representations that preserve row–column relationships, enabling

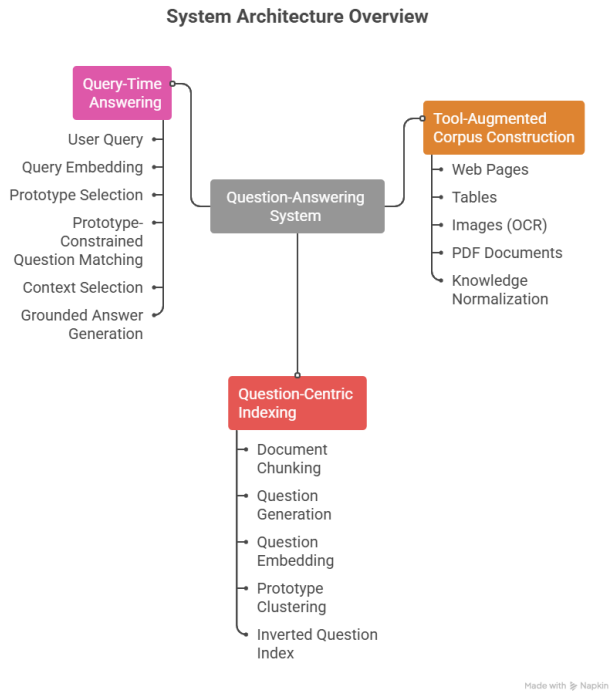

Fig. 1. Overview of the proposed tool-augmented QuIM-RAG system.

their integration into downstream processing. Image-based information is extracted using optical character recognition (OCR), with preprocessing techniques such as noise reduction and thresholding applied to improve text recognition quality. PDF documents are parsed to recover their textual content.

All extracted information is normalized into a unified textual representation, forming a consolidated corpus for indexing. This design enables the system to operate effectively on heterogeneous web-based knowledge sources beyond clean, text-only datasets.

B. Question-Centric Knowledge Indexing

In the second stage, the constructed corpus is processed using the question-centric indexing paradigm introduced in QuIM-RAG. Documents are segmented into overlapping chunks to maintain semantic coherence while respecting context length constraints. For each chunk, an instruction-tuned language model generates representative questions that capture its informational content.

These generated questions serve as the primary retrieval units, replacing raw document chunks. Each question is embedded into a shared semantic vector space, allowing both document-derived questions and user queries to be represented uniformly. To improve retrieval efficiency and reduce information dilution, question embeddings are clustered using centroid-based clustering. The resulting cluster centroids, referred to as prototypes, act as coarse semantic representatives of question groups.

An inverted index maps each prototype to its associated questions and corresponding document chunks, enabling ef-

ficient narrowing of the search space during query-time retrieval.

C. Query-Time Retrieval and Answer Generation

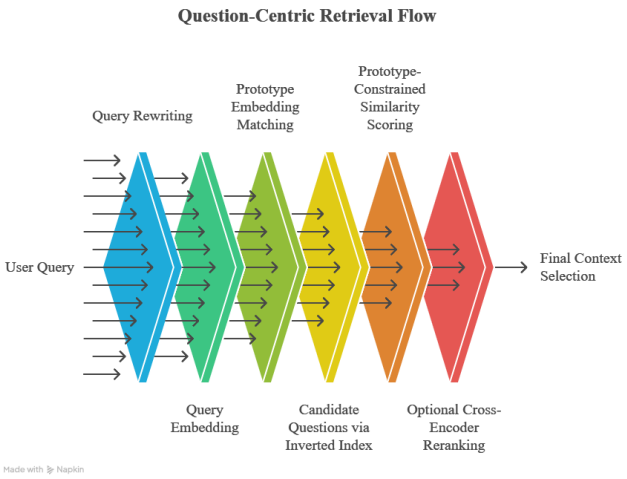

Fig. 2. Question-centric retrieval and prototype-constrained matching process.

During inference, a user query is optionally reformulated using a query rewriting module to improve clarity and retrieval effectiveness. The query is then embedded into the same semantic space as the indexed questions. The query embedding is compared against prototype embeddings, and the nearest prototype is selected to identify a relevant subset of candidate questions.

Within this reduced candidate set, dense similarity matching is performed to retrieve the most relevant questions and their associated document chunks. An optional cross-encoder reranking step further refines ranking precision. The top-ranked chunks are then provided as context to a language model, which generates responses grounded in retrieved evidence.

If no sufficiently relevant context is identified, the system abstains from answering, thereby reducing unsupported or hallucinated outputs. This controlled generation mechanism improves reliability in domain-specific question answering scenarios.

D. Design Rationale

The system architecture is designed to separate data acquisition, indexing, and retrieval into distinct modules. Tool-augmented corpus construction addresses the limitations of text-only ingestion, while question-centric indexing improves semantic alignment and reduces information dilution. Prototype-based indexing enables efficient coarse-to-fine retrieval, and auxiliary components such as query rewriting and reranking enhance robustness without modifying the core retrieval paradigm. Together, these design choices extend QuIM-RAG toward practical deployment in real-world environments.

IV. Methodology

This section describes the proposed methodology, detailing how tool-augmented corpus construction, question-centric indexing, prototype-based retrieval, and grounded answer generation are integrated into a unified system. The methodology extends the QuIM-RAG framework with data-centric and system-level enhancements designed for real-world web data and domain-specific deployments.

A. Tool-Augmented Web Corpus Construction

The first stage focuses on constructing a domain-specific knowledge corpus from real-world web sources. Unlike conventional RAG pipelines that assume pre-processed textual datasets, the proposed approach performs automated data acquisition and knowledge extraction from heterogeneous web artifacts.

Given a target domain, a crawler retrieves web pages under domain and depth constraints. For each page, multiple extraction techniques are applied: visible textual content is extracted after removing boilerplate elements, structured HTML tables are parsed and converted into textual representations preserving row–column semantics, embedded Image-based textual information is extracted using optical character recognition (OCR) with the Tesseract engine. To improve recognition accuracy under real-world conditions, preprocessing steps including grayscale conversion, noise reduction, and thresholding are applied prior to OCR. The extracted text is further cleaned and normalized before integration into the unified corpus.

TABLE I
Web Knowledge Extraction Modules

Content Type	Extraction Mechanism
Text	HTML parsing and boilerplate removal
Tables	Structured HTML table parsing
Images	OCR-based text extraction
PDFs	Document text parsing

B. Knowledge Chunking and Question Generation

The normalized corpus is segmented into overlapping chunks to maintain semantic coherence while respecting the context limits of language models. Let a document D be partitioned into chunks $\{c_1, c_2, \ldots, c_n\}$, where each chunk contains a fixed number of tokens with controlled overlap.

For each chunk c_i, an instruction-tuned language model generates a set of representative questions $\{q_{i1}, q_{i2}, \ldots, q_{ik}\}$ such that each question is answerable solely from the content of c_i. This process transforms passive knowledge into a query-aligned representation space.

Chunks that do not yield valid questions are excluded from indexing, ensuring that each indexed unit is associated with at least one explicit question.

C. Question Embedding and Prototype-Based Index Construction

Each generated question q is embedded into a shared semantic vector space using a sentence embedding model:

$$e_q = f_{\text{embed}}(q), \quad e_q \in \mathbb{R}^d$$

where $f_{\text{embed}}(\cdot)$ denotes the embedding function.

To improve retrieval efficiency and reduce information dilution, question embeddings are clustered using centroid-based clustering. Let Q denote the set of all questions. Clustering produces K prototype vectors $\{p_1, p_2, \ldots, p_K\}$, where each prototype is defined as:

$$p_k = \frac{1}{|C_k|} \sum_{q \in C_k} e_q$$

where C_k denotes the set of questions assigned to prototype p_k.

An inverted index is constructed that maps each prototype to its associated questions and corresponding document chunks, enabling efficient narrowing of the search space during query-time retrieval.

D. Query-Time Question-Centric Retrieval

At inference time, a user query u is optionally reformulated using a query rewriting module to improve clarity. The query is embedded into the same semantic space:

$$e_u = f_{\text{embed}}(u)$$

The query embedding is compared against prototype embeddings, and the nearest prototype p^* is selected using cosine similarity:

$$p^* = \arg\max_{p_k} \text{cosine}(e_u, p_k)$$

Candidate questions associated with p^* are retrieved from the inverted index. Within this restricted set, dense similarity matching is performed between the query embedding and candidate question embeddings. An optional cross-encoder reranking step further refines ranking precision. The top-ranked questions determine the corresponding document chunks used as retrieval context.

E. Grounded Answer Generation and Refusal Mechanism

As illustrated in Fig. 3, the system enforces a decision mechanism that ensures answers are generated only when sufficient supporting context is available.

The selected context is provided to a language model, which generates responses conditioned strictly on the retrieved evidence. If no sufficiently relevant context is identified, the system abstains from answering and returns a refusal response.

This design enforces strict grounding and reduces hallucination by ensuring that generated answers are supported by retrieved knowledge.

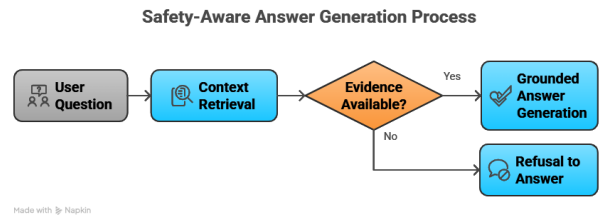

Fig. 3. Grounded answer generation with refusal mechanism. The system generates responses only when sufficient evidence is retrieved, otherwise abstains from answering.

F. Methodology Summary

The proposed methodology integrates tool-augmented corpus construction, question-centric indexing, prototype-based inverted retrieval, and grounded answer generation into a unified framework. By extending QuIM-RAG with data-centric and system-level enhancements, the approach enables effective operation over heterogeneous web-based knowledge sources.

Unlike conventional RAG pipelines that rely on clean textual corpora, the proposed system incorporates automated extraction from diverse data modalities, including structured tables, images, and PDF documents, thereby improving knowledge coverage in real-world settings. The adoption of question-centric indexing preserves semantic alignment between queries and indexed knowledge, while prototype-based clustering enables efficient coarse-to-fine retrieval.

At query time, retrieval is constrained to semantically relevant regions of the question space, followed by similarity matching and optional reranking to improve precision. The integration of a refusal mechanism further enforces strict grounding by preventing unsupported answer generation.

Overall, the methodology demonstrates how system-level design and data-centric enhancements can extend existing question-centric RAG frameworks toward reliable and practical domain-specific deployment.

TABLE II
COMPARISON OF RAG, QUIM-RAG, AND PROPOSED SYSTEM

Feature	RAG	QuIM-RAG	Proposed
Retrieval Unit	Chunks	Chunks	Chunks & Questions
Indexing	Dense/Hybrid	Prototype	Prototype
Data Type	Text	Text	Multi-modal
Corpus	Preprocessed	Preprocessed	Tool-Augmented + Preprocessed
Retrieval	Global	Prototype	Refined
Grounding	Limited	Moderate	Refusal-based
Deployment	Low	Medium	High

V. EXPERIMENTAL SETUP

This section describes the experimental configuration used to evaluate the proposed tool-augmented QuIM-RAG framework. We detail the dataset construction, model components, indexing strategy, retrieval pipeline, baseline systems, and reproducibility settings to enable independent replication.

A. Dataset and Corpus Construction

The experimental corpus is constructed using the tool-augmented web extraction pipeline described in Section IV-A. The dataset is derived from institutional and educational web sources and includes heterogeneous knowledge artifacts such as narrative text, structured HTML tables, image-embedded textual content extracted via optical character recognition (OCR), and linked PDF documents.

All extracted content is normalized into a unified textual representation. Boilerplate elements such as headers, navigation menus, and footers are removed to retain semantically meaningful information. The normalized documents are segmented into overlapping chunks to preserve semantic continuity while maintaining compatibility with downstream language model context limits.

TABLE III
CORPUS STATISTICS

Component	Quantity
Web pages crawled	∼40
Normalized documents	∼40
Knowledge chunks	∼108
Generated questions	∼615
Prototype clusters	256
Avg. questions per chunk	∼5–6

The resulting dataset represents a medium-scale, domain-specific corpus that reflects realistic institutional deployment scenarios. While limited in size, it is sufficient to evaluate system behavior and validate design choices in practical settings.

B. Question Generation Configuration

Each knowledge chunk is processed using an instruction-tuned language model to generate representative questions. Prompt design ensures that generated questions satisfy the following constraints:

- Each question is answerable solely using the corresponding chunk
- External knowledge or inference beyond the chunk is avoided
- Redundant, vague, or speculative questions are filtered

Only chunks that produce at least one valid question are retained. This filtering step ensures that each indexed unit is explicitly associated with answerable queries, supporting grounded retrieval and answer generation.

C. Embedding Model and Vector Representation

All generated questions, document chunks, and user queries are embedded into a shared semantic vector space using a sentence-level transformer model.

- Embedding model: `BAAI/bge-large-en-v1.5`
- Embedding dimensionality: 1024
- Vector normalization: L2 normalization
- Similarity function: Cosine similarity

Using a unified embedding model ensures consistency across indexing and retrieval stages.

D. Prototype-Based Index Construction

Let $Q = \{q_1, q_2, \ldots, q_n\}$ denote the set of generated questions, and e_{q_i} their embeddings. To reduce retrieval complexity and mitigate information dilution, embeddings are clustered using centroid-based clustering.

The number of clusters K (prototypes) is selected based on the size of Q. Each prototype is defined as:

$$p_k = \frac{1}{|C_k|} \sum_{q \in C_k} e_q$$

where C_k is the set of questions assigned to prototype p_k.

An inverted index maps each prototype to its associated questions and corresponding document chunks, enabling efficient narrowing of the retrieval space during query-time inference.

E. Vector Database Backend

All embeddings and associated metadata are stored using a vector database to support approximate nearest neighbor (ANN) search.

- Vector database: ChromaDB
- ANN method: HNSW-based indexing
- Stored metadata: question text, chunk identifier, source URL

Prototype clustering and index construction are performed offline and updated incrementally when new data is ingested.

F. Query-Time Retrieval Components

This subsection describes the query-time retrieval pipeline, which defines the core inference behavior of the proposed system.

1) Query Rewriting: User queries often exhibit ambiguity or underspecification. To improve retrieval robustness, a lightweight query rewriting module is optionally applied to reformulate the query into a clearer and self-contained form without altering its semantic intent. The rewritten query is used for retrieval, while the original query is preserved for answer generation. This component is treated as an auxiliary enhancement rather than a core methodological contribution.

2) Query Embedding: The (rewritten) query u is embedded using the same embedding model employed during indexing:

$$e_u = f_{\text{embed}}(u), \quad e_u \in \mathbb{R}^{1024}$$

Embedding normalization ensures consistent cosine similarity computation across all retrieval stages.

3) Prototype Selection via Query–Prototype Matching: To perform coarse-grained semantic routing, the query embedding e_u is compared against all prototype embeddings $\{p_1, p_2, \ldots, p_K\}$. The nearest prototype is selected as:

$$p^* = \arg\max_{p_k} \cos(e_u, p_k)$$

This step identifies the most relevant semantic region, reducing the retrieval scope from the entire corpus to a focused candidate subset.

4) Candidate Question Retrieval: Using the selected prototype p^*, the inverted index retrieves the associated set of candidate questions Q_{p^*}. Each candidate question is linked to corresponding document chunks. This step serves as a symbolic filtering mechanism that constrains retrieval to semantically aligned candidates.

5) Dense Similarity Matching: Within the candidate set Q_{p^*}, dense similarity matching is performed between the query embedding and candidate question embeddings:

$$\text{score}(q) = \cos(e_u, e_q)$$

Candidate questions are ranked based on similarity scores, enabling fine-grained retrieval refinement.

6) Cross-Encoder Reranking (Optional): To further improve ranking precision, a cross-encoder reranker is optionally applied to the top-k candidates.

- Reranker model: `cross-encoder/ms-marco-MiniLM-L-6-v2`

The cross-encoder jointly encodes the query and candidate question, capturing deeper semantic interactions beyond embedding-based similarity.

7) Context Selection: The highest-ranked candidate questions determine the corresponding document chunks used as retrieval context. Duplicate chunks are removed, and the total context length is constrained to fit within the input limits of the answer generation model.

G. Answer Generation and Refusal Mechanism

The retrieved context is provided to an instruction-tuned language model for response generation. The generation process enforces strict grounding by constraining the model to use only the provided context and avoid unsupported or speculative content.

If no sufficiently relevant context is identified, the system abstains from answering and returns a refusal response. This mechanism reduces hallucination and improves reliability in domain-specific question answering.

H. Baseline Systems

The proposed method is compared against the following baselines:

TABLE IV
BASELINE METHODS

Method	Description
BM25	Lexical retrieval using TF-IDF scoring
Dense	Chunk-centric dense retrieval
Hybrid	Combined lexical and dense retrieval
QuIM-RAG	Question-centric prototype-based retrieval
Proposed	Extended QuIM-RAG (tool-augmented system)

All methods use the same embedding models, generation models, and prompts to ensure fair comparison.

I. Hardware and Runtime Environment

Experiments were conducted using a cloud-based environment with GPU acceleration.

- Platform: Kaggle Notebooks
- GPU: $\sim$16 GB GPU memory
- CPU: Multi-core processor
- RAM: $\geq$32 GB

This setup reflects a practical and accessible computational environment rather than specialized high-performance infrastructure.

J. Reproducibility Protocol

All experiments follow a fixed configuration, including deterministic clustering parameters, controlled random seeds, and consistent prompt templates. The modular pipeline design supports reproducibility across corpus construction, indexing, retrieval, and answer generation stages.

VI. EVALUATION METRICS

Evaluating Retrieval-Augmented Generation (RAG) systems requires metrics that assess not only semantic correctness but also grounding, relevance, and safety. Traditional n-gram overlap metrics are insufficient, as they fail to capture factual consistency and dependence on retrieved evidence. Therefore, we adopt a comprehensive evaluation framework combining semantic, retrieval-aware, and safety-oriented metrics.

A. Semantic Answer Quality (BERTScore)

To measure semantic similarity between generated answers and reference answers, we use BERTScore, which computes token-level similarity in a contextual embedding space.

Given a generated answer A_g and a reference answer A_r, BERTScore computes precision (P), recall (R), and F1 score based on cosine similarity between token embeddings:

$$P = \frac{1}{|A_g|} \sum_{t_g \in A_g} \max_{t_r \in A_r} \cos(e_{t_g}, e_{t_r})$$

$$R = \frac{1}{|A_r|} \sum_{t_r \in A_r} \max_{t_g \in A_g} \cos(e_{t_r}, e_{t_g})$$

$$F_1 = \frac{2PR}{P+R}$$

BERTScore is well suited for generative QA tasks, as it captures semantic equivalence beyond exact token matching.

B. Faithfulness

Faithfulness measures the extent to which a generated answer is supported by retrieved context rather than hallucinated.

It is computed by:

- Decomposing the generated answer into atomic factual claims
- Verifying each claim against retrieved context
- Measuring the fraction of supported claims

Formally, if N denotes total claims and N_v denotes verified claims:

$$\text{Faithfulness} = \frac{N_v}{N}$$

Higher values indicate stronger grounding and reduced hallucination.

C. Answer Relevance

Answer relevance evaluates how well the generated answer addresses the original query.

Let q denote the query and $\{\hat{q}_1, \hat{q}_2, \ldots, \hat{q}_m\}$ denote questions generated from the answer. Relevance is defined as:

$$AR = \frac{1}{m} \sum_{i=1}^{m} \cos(e_q, e_{\hat{q}_i})$$

This metric captures alignment between answer content and query intent.

D. Context Precision

Context precision measures how much of the retrieved context is actually relevant for answering the query.

Let S denote all retrieved sentences and $S_r \subseteq S$ denote relevant sentences:

$$\text{Context Precision} = \frac{|S_r|}{|S|}$$

Higher values indicate more focused retrieval with less information dilution.

E. Harmfulness

To evaluate safety, generated answers are analyzed for harmful or unsafe content. If H denotes harmful claims and N denotes total claims:

$$\text{Harmfulness} = \frac{H}{N}$$

Lower values indicate safer and more controlled generation.

F. Evaluation Strategy

Two evaluation strategies are employed:

1) **Baseline Comparison**: A subset of queries is evaluated across BM25, dense, hybrid, and QuIM-RAG retrieval methods.
2) **Full System Evaluation**: A larger set of domain-specific queries is evaluated using the proposed system to assess grounding, relevance, and safety.

All evaluations are conducted under identical generation settings to isolate the effect of retrieval strategy.

G. Summary of Metrics

This combination of metrics provides a comprehensive evaluation of both answer quality and reliability in RAG systems.

VII. Results and Discussion

This section presents the experimental results of the proposed tool-augmented QuIM-RAG framework and analyzes retrieval effectiveness, answer quality, grounding, and safety. The proposed system is compared against lexical, dense, hybrid, and QuIM-RAG baselines.

TABLE V
Evaluation Metrics Summary

Metric	Objective
BERTScore	Semantic answer correctness
Faithfulness	Grounding in retrieved context
Answer Relevance	Query–answer alignment
Context Precision	Retrieval efficiency
Harmfulness	Safety and appropriateness

A. Baseline Comparison Results

We first evaluate retrieval performance using a representative subset of question–answer pairs. Table VI summarizes BERTScore precision, recall, and F1 scores for each retrieval strategy.

TABLE VI
Baseline Retrieval Comparison (BERTScore)

Method	Precision	Recall	F1
BM25	0.845	0.929	0.885
Dense	0.860	0.955	0.905
Hybrid	0.857	0.956	0.903
QuIM-RAG	0.853	0.951	0.899
Proposed	0.868	0.950	0.907

As shown in Table VI, dense and hybrid retrieval methods achieve high recall, indicating their ability to retrieve a broad set of potentially relevant content. However, this often comes at the cost of lower precision, as loosely related or redundant context may be included, which can negatively affect answer generation.

QuIM-RAG improves retrieval alignment by adopting question-centric indexing, resulting in more focused context selection compared to chunk-based retrieval methods. Building upon this, the proposed system introduces tool-augmented corpus construction and retrieval refinement mechanisms that further enhance the quality and relevance of retrieved context.

Although the improvements in BERTScore are moderate, the proposed system achieves competitive performance while demonstrating more consistent retrieval behavior. In particular, the improvements observed in grounding-oriented metrics such as faithfulness and context precision indicate that the proposed approach produces more reliable and contextually supported answers. This highlights the effectiveness of integrating data-centric enhancements with question-centric retrieval for practical deployment scenarios.

VIII. Scalability Considerations

This section discusses the scalability characteristics of the proposed system and provides a conceptual analysis of how question-centric indexing and prototype-based retrieval influence retrieval efficiency as corpus size increases.

A. Limitations of Chunk-Centric Retrieval

In conventional chunk-centric Retrieval-Augmented Generation (RAG) systems, retrieval is performed by comparing a query embedding against embeddings of all document chunks in the corpus. Let N denote the number of chunks and d the embedding dimensionality. Even with approximate nearest neighbor (ANN) methods, retrieval cost and memory usage grow with N.

As corpus size increases, two challenges become prominent:

- **Retrieval Cost Growth:** Retrieval latency increases with the number of indexed chunks, requiring more efficient search strategies.
- **Information Dilution:** Larger corpora increase the likelihood of retrieving partially relevant or noisy contexts, which can negatively impact answer generation.

These challenges motivate retrieval strategies that reduce the effective search space while maintaining semantic relevance.

B. Effect of Question-Centric Indexing and Prototype-Based Retrieval

The proposed system addresses these challenges by adopting question-centric indexing and prototype-based clustering, as introduced in QuIM-RAG. By organizing semantically similar questions into prototype groups, retrieval is constrained to a smaller subset of the search space.

At query time, the system first identifies the most relevant prototype and then performs similarity matching within that subset, effectively reducing the number of candidate comparisons. This coarse-to-fine retrieval strategy improves efficiency while preserving semantic alignment.

Although the current evaluation is conducted on a medium-scale corpus, the design of the retrieval pipeline suggests improved scalability characteristics compared to global chunk-based retrieval approaches. Future work will evaluate performance on larger-scale datasets.

C. Impact on Retrieval Latency and Memory Usage

By constraining retrieval to a semantically focused subset of questions, the proposed system reduces the number of candidate comparisons required during inference. As a result, dense similarity matching is performed over a smaller set of embeddings, which can improve retrieval efficiency compared to global chunk-based retrieval.

This design also promotes better utilization of vector database operations, as fewer embeddings are accessed per query. While the current evaluation is conducted on a medium-scale corpus, these characteristics suggest potential improvements in retrieval latency and resource usage as corpus size increases.

D. Scalability of Tool-Augmented Corpus Construction

The tool-augmented data acquisition pipeline operates as an offline preprocessing stage. Text extraction, table parsing, OCR, and PDF processing are performed once per document and do not directly impact query-time latency.

Although integrating heterogeneous data sources increases corpus richness, the use of question-centric indexing and prototype clustering abstracts the underlying content into a structured representation. This allows the system to incorporate diverse knowledge types without proportionally increasing retrieval complexity.

E. Incremental Index Updates

The proposed system supports incremental updates, which are common in practical deployment scenarios. When new documents are added:

1) New chunks are generated
2) Questions are produced and embedded
3) Embeddings are assigned to the nearest existing prototype
4) The inverted index is updated accordingly

This approach avoids full re-indexing of the corpus. Prototype re-computation may be performed periodically but is not required for every update.

F. Scalability Trade-offs

Prototype-based retrieval introduces additional offline computation, including question generation and clustering. However, these costs are amortized during indexing and enable more efficient query-time retrieval.

Compared to dense and hybrid baselines, the proposed system trades modest preprocessing overhead for improved retrieval focus and efficiency during inference.

G. Summary

Overall, the proposed framework reduces the effective retrieval search space through question-centric indexing and prototype-based clustering. While the current evaluation is limited to a medium-scale corpus, the design indicates favorable scalability characteristics for larger and more heterogeneous datasets. Future work will evaluate performance under large-scale conditions.

IX. Limitations

While the proposed system demonstrates improved grounding and practical applicability, several limitations remain.

A. Dependence on Question Generation

The effectiveness of question-centric retrieval depends on the quality of generated questions. Incomplete or generic questions may lead to reduced coverage of the underlying knowledge, affecting retrieval performance.

B. Noise in Extracted Content

The corpus construction pipeline relies on OCR and document parsing for images and PDFs, which may introduce noise due to low-quality inputs or complex layouts. Such inaccuracies can propagate to downstream retrieval and generation stages.

C. Evaluation Scale

Experiments are conducted on a medium-scale domain-specific dataset. While sufficient to demonstrate system behavior, the evaluation does not reflect large-scale deployment scenarios, and scalability conclusions remain indicative rather than empirically validated.

D. Clustering Sensitivity

Prototype-based indexing depends on clustering quality and parameter selection. Suboptimal clustering may lead to uneven distribution of questions or overlapping semantic regions, affecting retrieval efficiency.

E. Preprocessing Overhead

The framework introduces additional offline computation, including question generation and clustering. Although this improves query-time efficiency, preprocessing cost may become significant for frequently updated corpora.

F. Language Scope

The current implementation focuses on English-only data and does not support multilingual or cross-lingual retrieval scenarios.

G. Evaluation Constraints

Evaluation relies primarily on automated metrics. While effective for large-scale analysis, these may not fully capture nuanced human judgments of answer quality.

H. Summary

These limitations highlight trade-offs between retrieval quality, preprocessing cost, and evaluation scope, and suggest directions for future improvements.

X. Conclusion and Future Work

In this work, we presented a tool-augmented question-centric retrieval framework that extends QuIM-RAG toward practical deployment in domain-specific knowledge environments. Unlike conventional RAG systems that rely on chunk-centric retrieval over text-only corpora, the proposed approach integrates heterogeneous web data—including structured tables, image-based text via OCR, and PDF documents—into a unified question-centric indexing pipeline.

By generating representative questions and organizing them into prototype-based clusters, the system enables coarse-to-fine retrieval that reduces the search space while maintaining semantic alignment with user queries. Experimental results demonstrate that, while dense retrieval achieves slightly higher recall, the proposed method provides competitive semantic performance along with improved grounding and reliability, reducing the likelihood of unsupported responses. Additionally, auxiliary components such as query rewriting and cross-encoder reranking enhance retrieval robustness without altering the core retrieval design.

The analysis indicates that prototype-based indexing can improve retrieval efficiency by constraining the search space, while the tool-augmented corpus construction pipeline enables the system to operate effectively on heterogeneous web-based data. These findings highlight the importance of combining question-centric retrieval with data-centric design for building reliable RAG systems in practical settings.

A. Future Work

Several directions can further enhance the proposed framework:

- **Multilingual Extension:** Extending the system to support multilingual and cross-lingual retrieval.
- **Adaptive Re-Clustering:** Dynamically updating prototypes under corpus evolution and domain shifts.
- **Robust Information Extraction:** Improving OCR and document parsing for complex layouts and noisy inputs.
- **Human-Centered Evaluation:** Incorporating human judgments of usefulness and trustworthiness.

Overall, this work demonstrates that extending question-centric retrieval frameworks with practical data acquisition and retrieval mechanisms can improve the reliability and applicability of RAG systems in real-world scenarios.

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